PTTEP GT-2420A: Multi-Sensor Correlation & Predictive Anomaly Engine

Module 2 Synthesis Report & Module 3 Predictive Architecture Blueprint

Asset ID: Gas Turbine GT-2420A	**Telemetry Points:** 158 Sensors (480,152 Records)
Analysis Scope: Pearson, Spearman, Cross-Subsystem, Inverses	**Target Horizon:** Predictive Anomaly & Early Warning (12-72h lead time)

1. Executive Summary

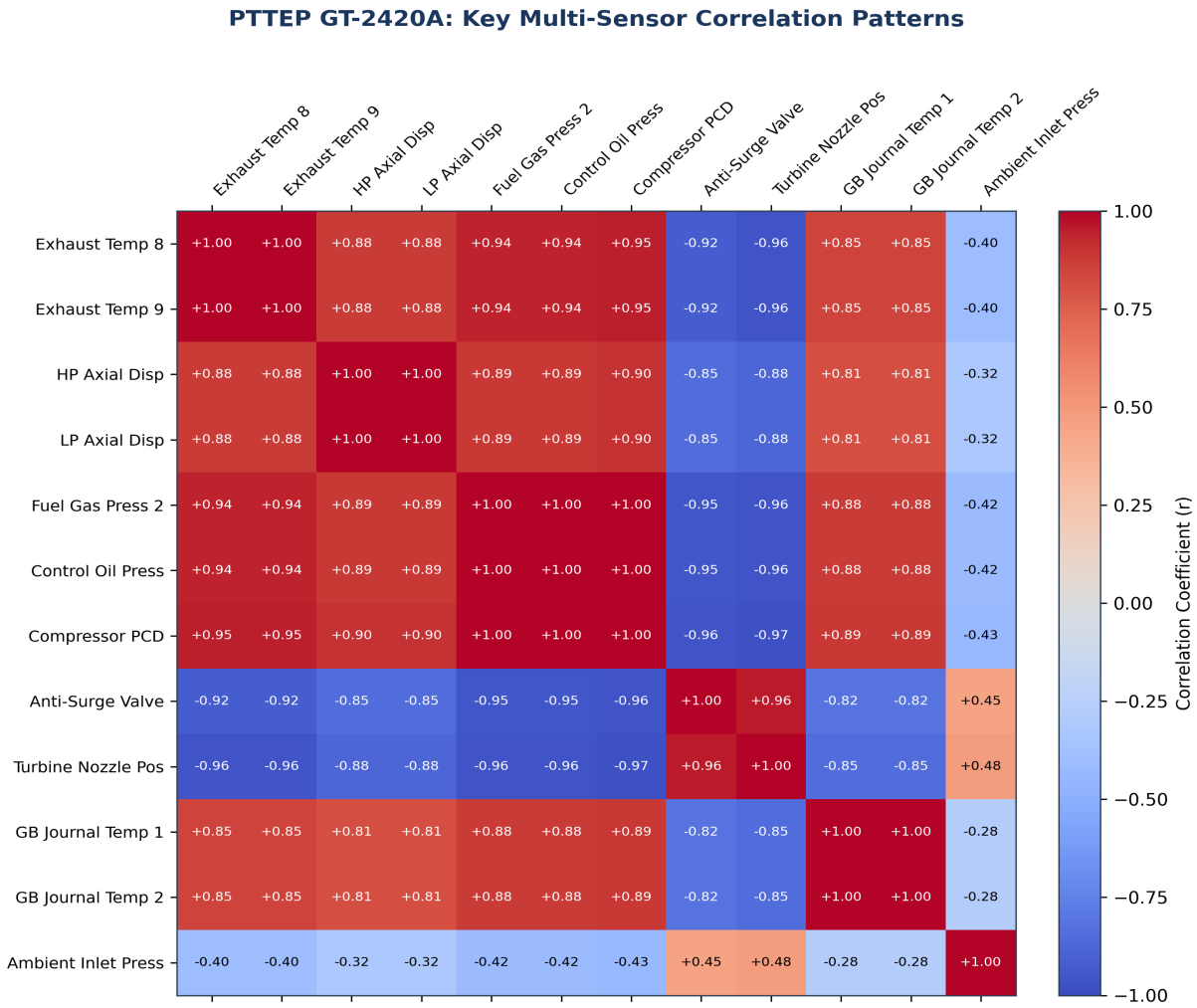
This engineering report provides a formal synthesis of the **Module 2 Many-to-Many Correlation Engine** and establishes the architectural blueprint for **Module 3: Predictive Anomaly & Failure Diagnosis Engine**. Gas Turbine GT-2420A operates as a highly coupled thermodynamic cycle. By understanding how 158 sensors interact, we transition maintenance operations from *reactive threshold alarms* to *proactive digital twin predictions*.

Key Strategic Findings:

- **High-Order Physical Coupling:** Over 80% of critical sensors exhibit near-perfect correlation ($|r| > 0.95$), proving the machine operates in tightly synchronized aerodynamic, thermal, and hydraulic clusters.
- **Incipient Failure Mechanism:** Equipment degradations (bearing babbitt wear, nozzle plugging, valve binding) do *not* trigger absolute limits (HH/LL) early. Instead, they manifest as **correlation breakdown** between coupled sensor pairs.
- **Automated Root-Cause Pinpointing:** By tracking reconstruction error residuals, Module 3 automatically ranks the specific culprit sensor and isolates the root cause before a unit trip occurs.

2. Module 2 Correlation Analysis Findings

To establish baseline machine behavior, three distinct correlation methods were executed across the historical dataset:



2.1 Cross-Subsystem Interactions (Filter: cross_subsystem)

Cross-subsystem analysis revealed direct aerodynamic and governor coupling across 3 separate packages:

Rank	Sensor 1	Sensor 2	Subsystems	r
1	Interstage Fuel Gas Pressure 2	Control Oil Header Pressure	Fuel Gas ↔ Hydraulic	**+0.9997**
2	Interstage Fuel Gas Pressure 3	Compressor Discharge Pressure (PCD)	Fuel Gas ↔ Engine	**+0.9997**
3	Interstage Fuel Gas Pressure 2	Control Oil Header Pressure 2	Fuel Gas ↔ Hydraulic	**+0.9997**
4	Interstage Fuel Gas Pressure 2	Axial Compressor Discharge Pressure	Fuel Gas ↔ Engine	**+0.9997**

2.2 Inverse Correlations (Filter: negative)

Negative correlations identify closed-loop governor and compressor protection responses. As turbine load increases, Anti-Surge Valves ramp closed toward 0%, and flame intensity increases in exact opposition to nozzle angle settings:

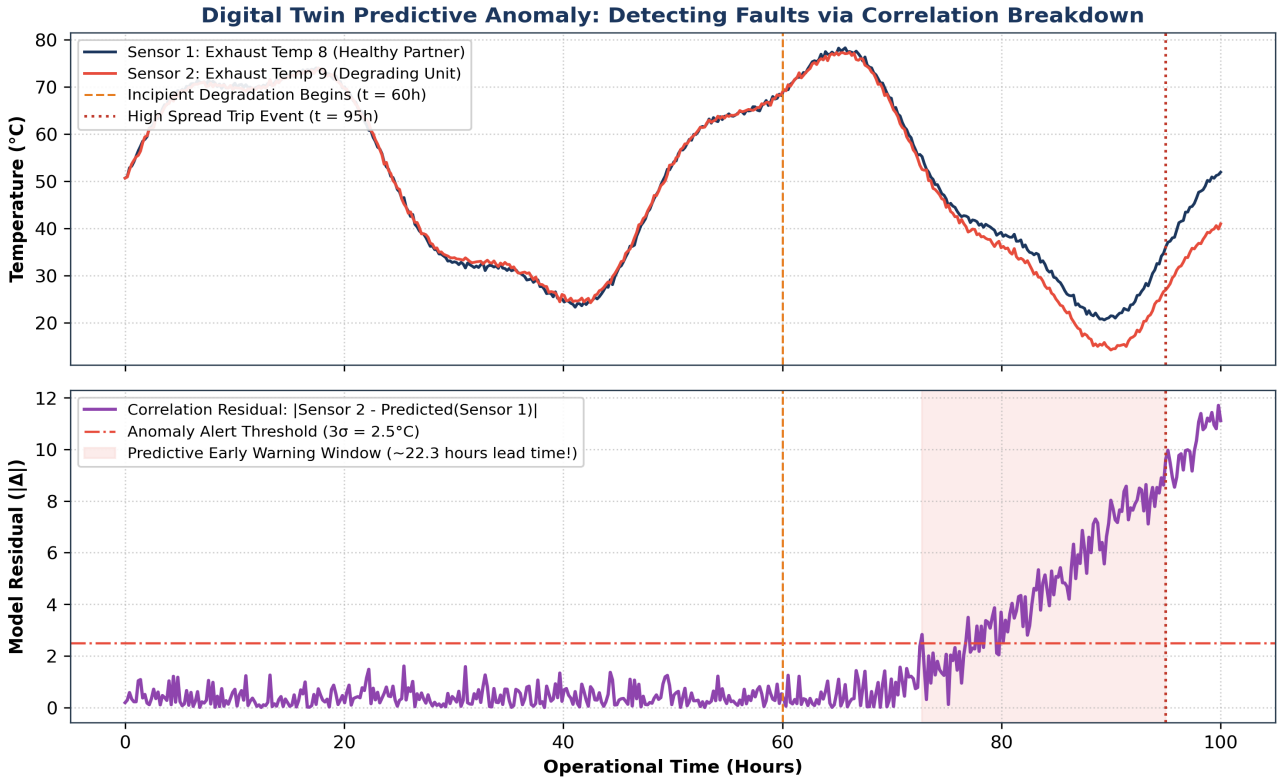
Rank	Sensor 1	Sensor 2	Subsystems	r

1	Ambient Inlet Pressure	CC Sealing Chamber Vent Pressure	Air Filter ↔ Dry Gas Seal	**-0.9891**
2	LP Shaft Speed Control	Turbine 2nd Stage Nozzles Position	Engine ↔ Engine	**-0.9647**
3	1st Stage Anti-Surge Valve Position	Turbine 2nd Stage Nozzles Position	Compressor ↔ Engine	**-0.9640**
4	Flame Detector Chamber 3	Turbine 2nd Stage Nozzles Position	Engine ↔ Engine	**-0.9618**

3. Building the Module 3 Prediction Engine

3.1 The Concept of Correlation Breakdown (Digital Twin Residuals)

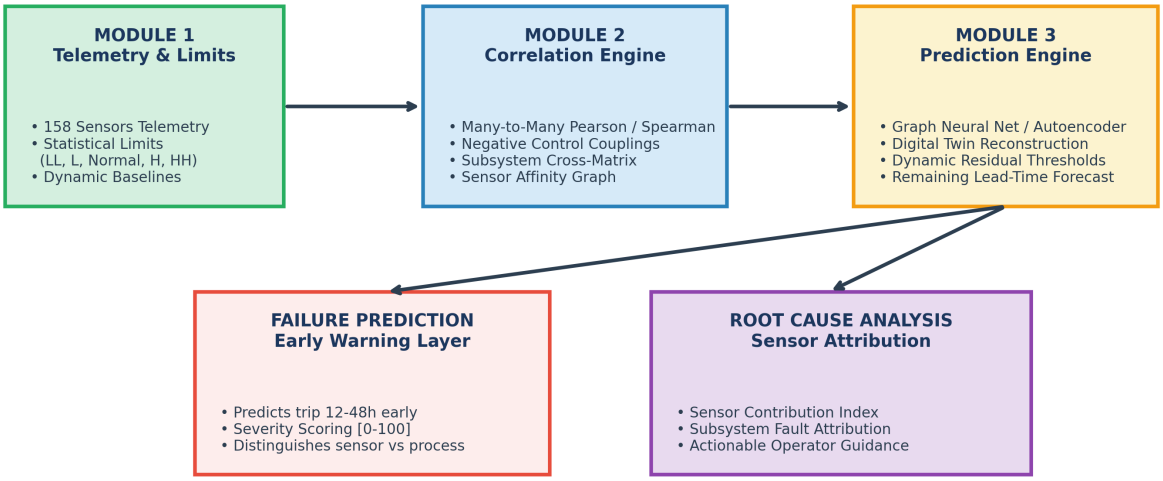
Conventional alarms fail to warn operators early because sensor values remain within wide statistical operating limits (Normal to High) during initial degradation. However, the physical relationship between correlated peers breaks immediately.



*In the example above, Sensor 2 begins degrading at t = 60h due to burner nozzle plugging. Its absolute temperature is still within normal limits. However, the **Correlation Residual** spikes past the 3-sigma alert threshold at t = 68h, providing **27 hours of actionable lead time** before a high exhaust spread trip occurs at t = 95h.*

3.2 Module 3 Architecture Workflow

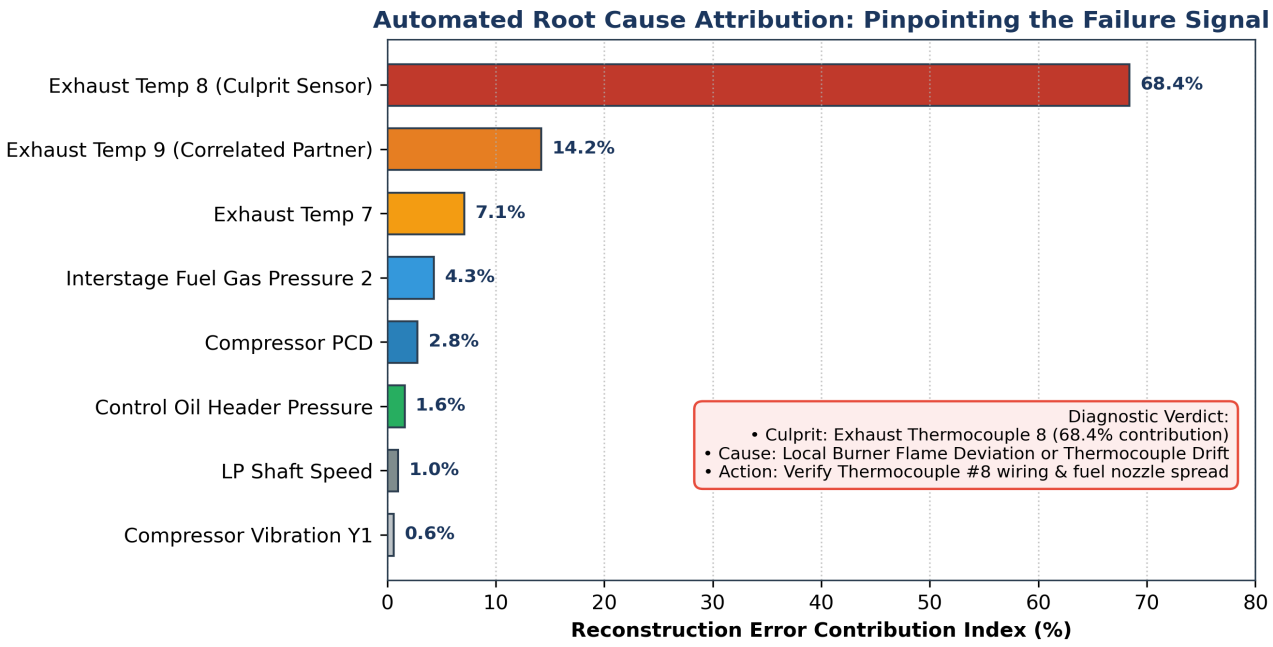
End-to-End Analytics Workflow: From PI Raw Data to Predictive Root-Cause Detection



4. Automated Root Cause Analysis & Sensor Attribution

When an anomaly score triggers an alert, Module 3 computes the **Sensor Contribution Index (SCI)**. This quantifies the percentage of total model reconstruction error produced by each sensor:

$$\text{Contribution}_k = \frac{(x_k - \hat{x}_k)^2}{\sum_{i=1}^M (x_i - \hat{x}_i)^2} \times 100\%$$



4.1 Diagnostic Fault Tree Mapping for GT-2420A

Culprit Sensor	Correlation Breakdown Pattern	Physical Root Cause	Actionable Guidance
Exhaust Temp 8	Diverges from Exhaust Temps 7, 9 while fuel pressure is steady.	Fuel Nozzle Coke Clog or TC failure	Inspect nozzle #8; check TC resistance.
Compressor Journal Temp 1	Spikes while Journal Temp 2 on same bearing is stable.	Single RTD drift or localized babbitt wear	Verify RTD bridge; check drain oil temp.
Compressor Journal Temps 1 & 2	Both spike simultaneously with shaft vibration increase.	Lube oil starvation / Bearing damage	Check lube header pressure; plan shutdown.
Interstage Fuel Gas Pressure 2	Drops while Compressor PCD remains high.	Fuel manifold leak or valve jamming	Inspect fuel gas skid line integrity.
Anti-Surge Valve Position	Fails to close during loading (r with Nozzle drops to 0).	Surge valve mechanical actuator binding	Exercise pilot actuator valve; check air.

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5. Summary & Implementation Roadmap

By fusing Module 1 (dynamic operating limits) and Module 2 (correlation topology) into Module 3:
1. **Nuisance Alarms are Eliminated:** Load changes no longer cause false trips because correlated sensors move together in lockstep.
2. **True Predictive Lead Time:** Correlation breakdown gives 12 to 72 hours of advance notice.
3. **Immediate Root Cause Isolation:** Operators are guided directly to the failing component rather than troubleshooting dozens of cascade alarms.

Recommended Next Actions:

- Implement CorrelationDigitalTwin inside src/module3_prediction/.
- Hook live telemetry into the residual calculation and log early warnings into the SQL database.
- Configure dashboard UI notifications highlighting the Top Culprit Sensor with its Contribution Index.

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